

BYTES AND BIRTHS

The Effect of Broadband Internet on Infant Health

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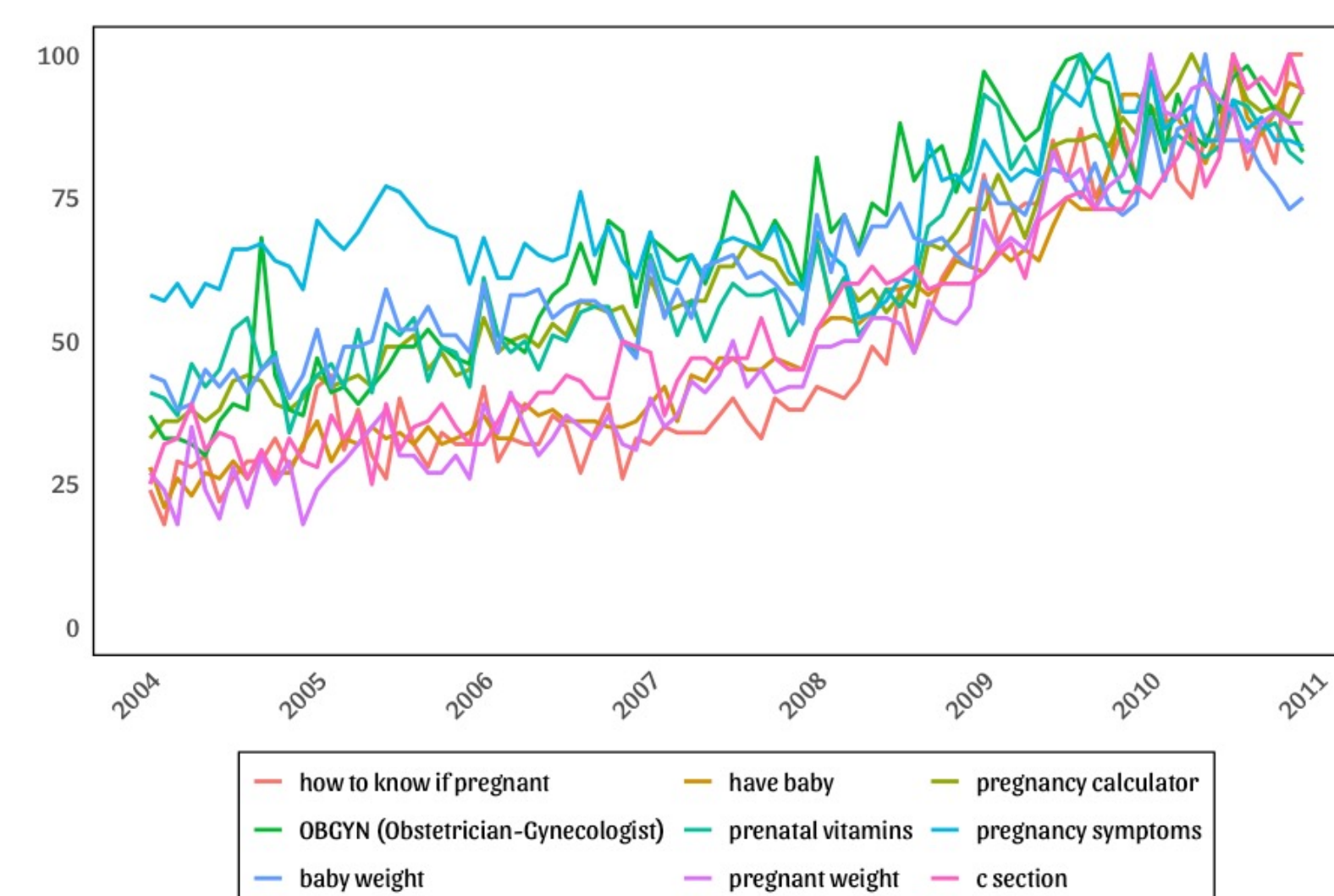


INTRODUCTION

From Books to Broadband

- Broadband lowers the marginal cost of getting health-related information

Google Trend for Pregnancy Related Words (2004-2010)



Information and Human Capital Formation

- Pregnancy women seek guidance (e.g., fetal development, nutrition, complications, and delivery preparation), and the health and well-being of newborns form a critical foundation for human capital development
- Reducing information friction within the medical community, thereby further altering practice patterns and the adoption of health technologies

The U.S. Context and Paradox

- Large disparities in maternal and infant health persists
- The U.S. trails behind other OECD countries with notably high infant mortality and low birthweight rates

RESEARCH QUESTION

How does better access to (online) health information affect infant health?

DATA & METHODS

Nativity Detail Files (2001-2008)

- Birth weight* as primary outcome due to its objective and precise measure

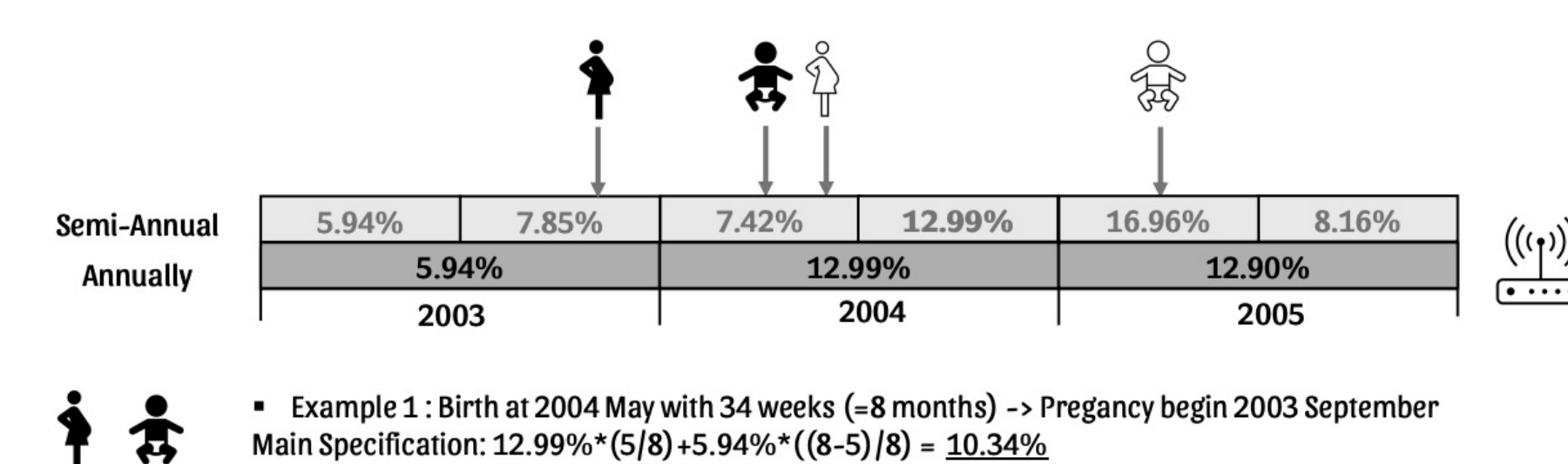
Federal Communications Commission (FCC) Form 477 data

- The number of Internet Service Providers (ISPs) within a zip code that serve at least one customer with broadband speeds of 200 kbps or higher
- Construct following *Detting et al. (2018)*, then aggregate to county level

$$1(\text{broadband})_z = 1(\text{per provider population} \leq 2700)_z \times 1(\text{per provider area} \leq 12 \text{ miles}^2)_z$$

- Duration-weighted treatment as considering the conception

Construct of Broadband Availability

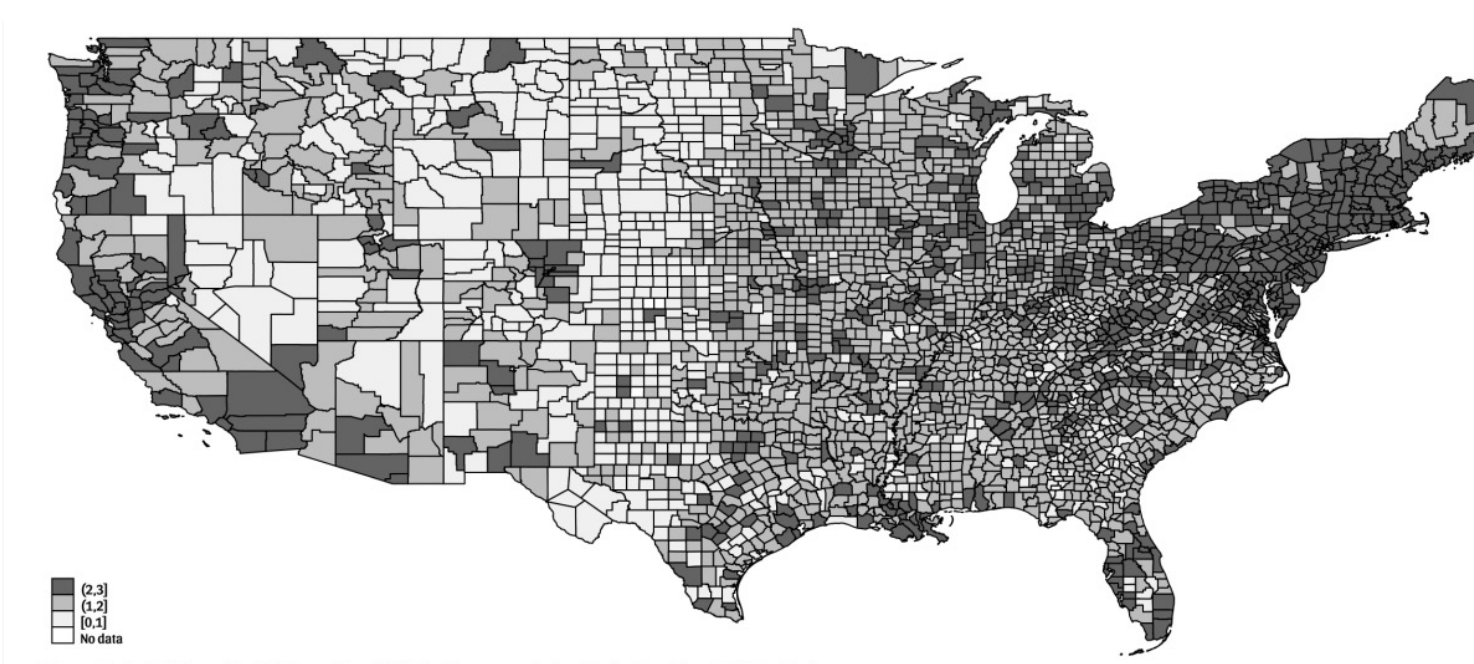


Staggered Difference-in-Differences

$$Y_{ct} = \alpha + \beta \text{Broadband}_{ct} + \phi X_{ct} + \delta_c + \gamma_t + \epsilon_{ct}$$

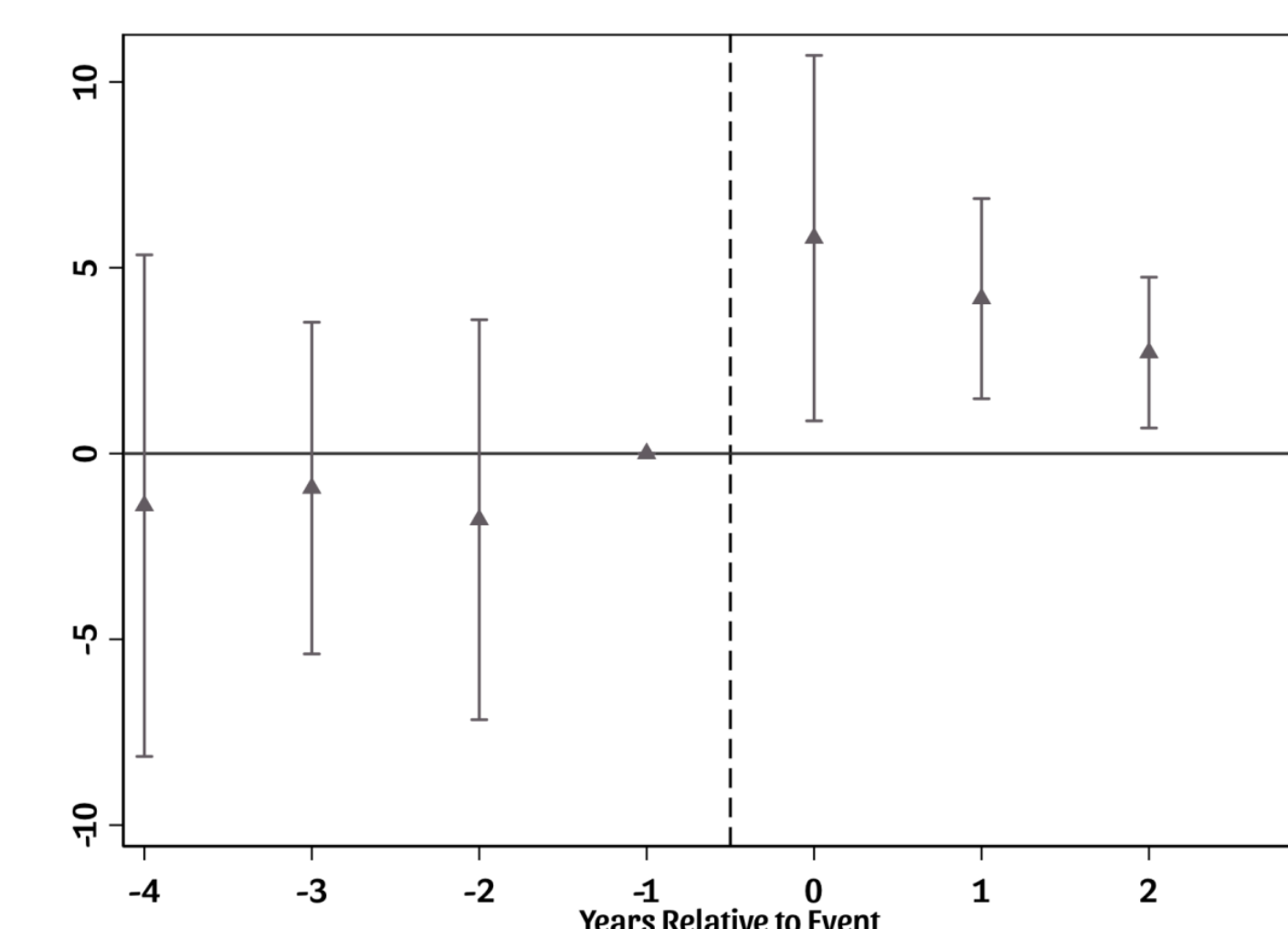
- Intent-to-treat (ITT)* effects of broadband availability from 0 to 1
- De Chaisemartin and d'Haultfoeuille
- 25% bin thresholds (quartiles) with 4 groups and 4 pre / post periods

Variation in Broadband Treatment by Counties (2000-2008)



RESULTS

Increase in average birth weight of 7.5 grams



- Reduce the incidence of *low-birth-weight* births (1,500-2,499g) by 3.6% while increase the incidence of *high birth weight* births (>4000g, 4.8%)
- Magnitudes are similar to the major U.S. *safety-net programs*, such as food stamps, WIC, and the FMLA

Where Does the Increase in Birth Weight Come from?

Fertility Selection into Birth

- Broadband leads to a reduction in the total birth rate of -1.72, about 2.37%
- Composition Change (e.g., fertility rises among Non-White and low educated with those having larger effects)

Longer Gestation and Pregnancies

- Effects disappear when controlling for gestational length, with no impact on fetal growth (e.g., SGA/LGA)

Behavioral channels

- Later initiation of prenatal care* may indicate broader access to information
- Complementary results from BRFSS shows the improvement of mental health and physical activity for Non-White

CONCLUSIONS

- Broadband availability raises birth weights and alters the distribution
- Digital access is an important determinant of shaping early-life human capital formation
- Policy implication: Pair broadband access with digital literacy to prevent algorithms from worsening health disparities

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Working Papers

- Fetal Origins of Mental Health: Evidence from Natural Disaster in Taiwan (& Elaine M. Liu, Jin-Tan Liu, and Tzu-Yin Hazel Tseng)
- Public Payment Mandates and Provider Supply (& Lauren Hoehn-Velasco, Olanrewaju Yusuff)
- Full Day Kindergarten Mandates and Child Maltreatment (& Bogyung Kim)

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